

BIOINFORMATICS

WEEK 2 – DAY 13

Mini Project 2: Disease Gene Analysis

STEP 1 : Chose a Disease Gene - TP53

STEP 2 : Know more about the gene

- **TP53** is a gene that codes for a protein called p53, which acts as the ultimate "master brake" and quality control inspector inside human cells.
- It is widely known in biology as the "guardian of the genome" because its primary job is to protect your DNA from mutating and becoming cancerous.

How it Works (The Cell's Emergency Responder)

Inside a healthy cell, the p53 protein constantly monitors DNA stability. When damage occurs, TP53 activates a three-step protocol:

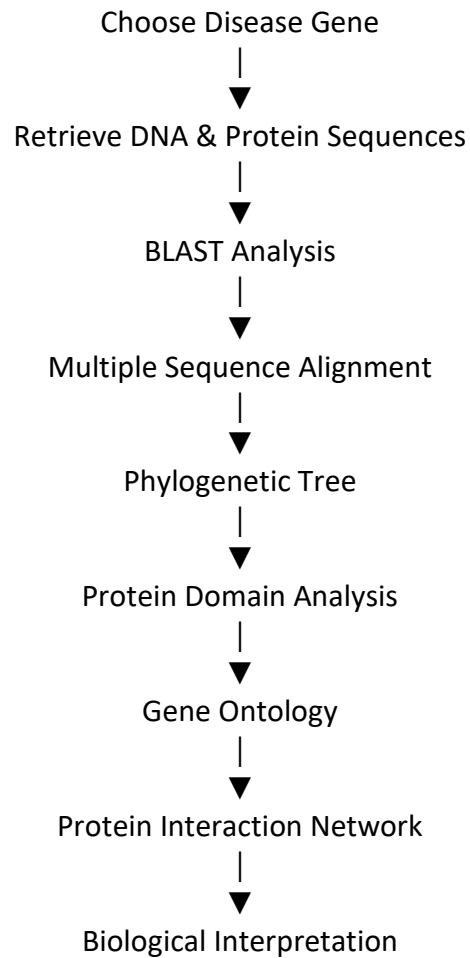
- **Stop (Cell Cycle Arrest):** It pauses cell division so repair crews have time to fix the broken DNA.
- **Fix (DNA Repair):** It activates specific genes responsible for mending genetic code errors.
- **Destroy (Apoptosis):** If the damage is too severe to fix, it orders the cell to self-destruct so the mutation cannot spread.

Real-World Connection to Cancer

Because TP53 stops damaged cells from multiplying, tumors cannot easily form while it is functioning normally.

- **The Most Common Mutation:** TP53 is the most frequently mutated gene across all human cancers.
- **Loss of Control:** In your Python exercise data, the cancer sample showed a drop from 50 down to 18 expression units. When TP53 expression drops or the gene is mutated, the cell loses its braking mechanism, allowing mutated cells to divide uncontrollably and form tumors.

STEP 3 : Project workflow :



STEP 4 – Retrieved Sequences

Download from NCBI or UniProt:

- TP53 DNA (FASTA)
- TP53 protein (FASTA)

Also retrieve TP53 protein sequences from:

- Human
- Chimpanzee
- Mouse
- Cow
- Zebrafish

Save in : Data/

STEP 5 – BLAST

Ran: BLASTn & BLASTp

STEP 6 – Multiple Sequence Alignment

Aligned the TP53 proteins.

Observed :

- Conserved regions.
- Variable regions.
- Gaps.

STEP 7 – Built a Phylogenetic Tree

Constructed a tree using the alignment.

STEP 8 – Protein Domains

Used InterPro to Identify:

- DNA-binding domain
- Tetramerization domain
- Regulatory regions (if shown)

STEP 9 – Gene Ontology (GO)

Gene Ontology provides standardized descriptions of gene function.

It has three categories:

GO Category	Meaning
Biological Process (BP)	What the gene does in the cell (e.g., apoptosis, DNA repair)
Molecular Function (MF)	The biochemical activity (e.g., DNA binding)
Cellular Component (CC)	Where the protein acts (e.g., nucleus)

STEP 10 – Protein Interaction Network

Used the STRING database to search: TP53

Observed interactions with proteins such as:

- MDM2
- ATM
- BRCA1
- CHEK2